

LECTURE 11 (153-248)

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2 \text{ReLU}(t-2) + \dots + \beta_{n-1} \text{ReLU}(t-(n-1)) + \varepsilon_t$$

$\text{ReLU}(x) = \max(x, 0)$

$$y_t = \beta_0 + \beta_1(t-1) + \varepsilon_t$$

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \varepsilon_t$$

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2(t-c_1)_+ + \beta_3(t-c_2)_+ + \varepsilon_t$$

Fit this model to the data with additional restrictions on $\beta_2, \dots, \beta_{n-1}$.

TREND

$$y_t = \mu_t + \varepsilon_t$$

$$\mu_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \dots + \beta_{n-1}(t-(n-1))_+$$

$$\beta_0 = \mu_1$$

$$\beta_1 = \mu_2 - \mu_1$$

$$\beta_2 = (\mu_3 - \mu_2) - (\mu_2 - \mu_1)$$

$$\beta_i = (\mu_{i+1} - \mu_i) - (\mu_i - \mu_{i-1})$$

We will use the restrictions:

$$\textcircled{1} \beta_2^2 + \dots + \beta_{n-1}^2 \leq \text{some quantity}$$

$$\textcircled{2} |\beta_2| + \dots + |\beta_{n-1}| \leq \text{some quantity.}$$

Least Squares

$$\min_{\beta_0, \beta_1, \dots, \beta_{n-1}} \sum_{t=1}^n \left[y_t - \beta_0 - \beta_1(t-1) - \beta_2(t-2)_+ - \dots - \beta_{n-1}(t-(n-1))_+ \right]^2$$

→ solution is $\beta_0 = y_1$

$$\beta_1 = y_2 - y_1$$

$$\beta_j = (y_{j+1} - y_j) - (y_j - y_{j-1})$$

$$\min_{\beta_0, \dots, \beta_{n-1}} \sum_{t=1}^n \left[y_t - \beta_0 - \beta_1(t-1) - \beta_2(t-2)_+ - \dots - \beta_{n-1}(t-(n-1))_+ \right]^2$$

→ RIDGE

$$\sum_{j=2}^{n-1} \beta_j^2 \leq V$$

$$\min_{\beta_0, \dots, \beta_{n-1}} \sum_{t=1}^n \left[y_t - \beta_0 - \beta_1(t-1) - \dots - \beta_{n-1}(t-(n-1))_+ \right]^2$$

→ LASSO.

$$\sum_{j=2}^{n-1} |\beta_j| \leq V$$

Practice

$$\min_{\beta_0, \dots, \beta_{n-1}} \sum_{t=1}^n \left[y_t - \beta_0 - \beta_1(t-1) - \dots - \beta_{n-1}(t-(n-1))_+ \right]^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2$$

$\min_{\beta_0, \beta_1, \dots, \beta_{n-1}} \sum (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_{n-1} x_{i,n-1})^2 + \lambda \sum_{j=1}^{n-1} |\beta_j|$
 $\rightarrow \hat{\beta}_{\text{LASSO}}(\lambda)$

y : real number

EXAMPLE

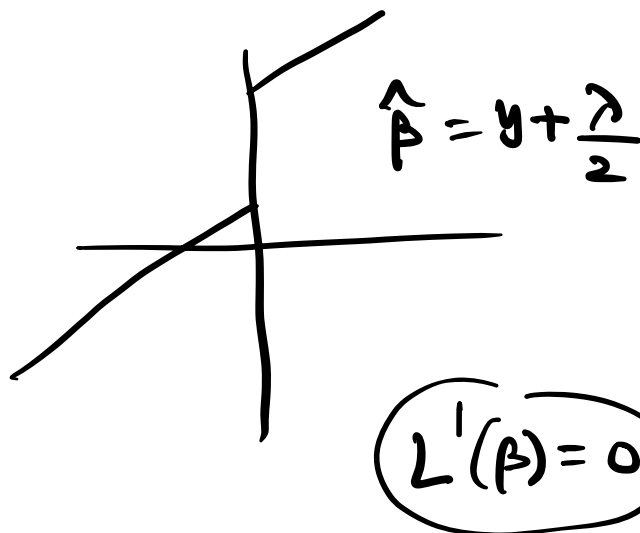
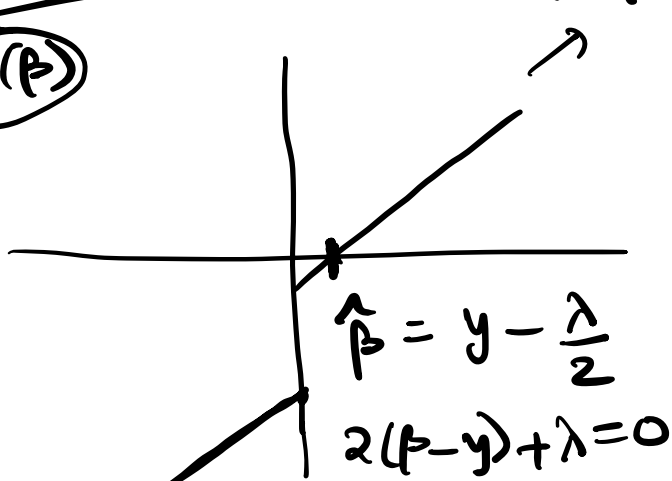
① $(y - \beta)^2 + \lambda \beta^2$
 minimize over β $\Rightarrow$ Differentiate: $2(\beta - y) + 2\lambda\beta = 0$
 $\Rightarrow \hat{\beta} = \frac{y}{1 + \lambda}$
 SHRINKAGE

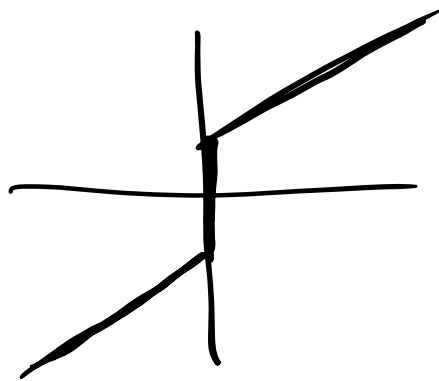
② $(y - \beta)^2 + \lambda |\beta| = L(\beta)$

Differentiate: $\begin{cases} 2(\beta - y) + \lambda & \text{if } \beta > 0 \\ 2(\beta - y) - \lambda & \text{if } \beta < 0 \end{cases}$

Case 1

$L'(\beta)$





$$\hat{\beta} = 0$$

Ans: $(y - \beta)^2 + \lambda |\beta|$

minimizer =
$$\begin{cases} y - \frac{\lambda}{2} & \text{if } y > \frac{\lambda}{2} \\ y + \frac{\lambda}{2} & \text{if } y < -\frac{\lambda}{2} \\ 0 & \text{if } -\frac{\lambda}{2} \leq y \leq \frac{\lambda}{2} \end{cases}$$

CHECK

CROSS-VALIDATION FOR PICKING λ

Split the times $1, \dots, n$ into two groups

T_{train} Δ T_{test}

80% 20%

Fix λ & fit

$$\min_{\beta} \left\{ \sum_{t \in T_{\text{train}}} \left[y_t - \beta_0 - \beta_1(t-1) - \dots - \beta_{n-1} \text{ReLU}(t-n) \right]^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2 \right\} \rightarrow \hat{\beta}(\lambda)$$

$$\hat{y}_t = \hat{\beta}_0(\lambda) + \hat{\beta}_1(\lambda)(t-1) + \hat{\beta}_2(\lambda)(t-2) + \dots + \hat{\beta}_{n-1}(\lambda)(t-(n-1))_t$$

for all $t \in T_{\text{test}}$

$$y_t$$

$$\text{Test Error}_{\text{Ridge}}(\lambda) = \sum_{t \in T_{\text{test}}} (y_t - \hat{y}_t)^2$$

$$\lambda = \{0.01, 0.1, 1, 10, 100, 1000\}$$

① 1st 80% → Train
last 20% → Test

② everything except 2nd 20% data → Train

1, 2, ..., n

① 2, 3, 4, 5, ⑥, 7, 8, 9, 10, ⑪, ...